



Scalability & Future Plan

Date	25 September 2024
Team ID	SWTID-2026-6721
Project Name	AI FAQ Assistant API
Maximum Marks	1 Mark

Scalability & Future Plan:

The AI FAQ Assistant API can be extended to support more users, larger FAQ collections, improved search capabilities, and additional AI-powered features. The future plan focuses on improving performance, security, deployment, and user experience.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The application currently runs primarily in a local development environment.	Limited accessibility for multiple users.	High
2	FAQ search currently uses keyword-based matching.	May not understand the meaning behind complex queries.	Medium
3	AI responses depend on the availability and limits of the Gemini API.	AI features may be affected by API availability or usage limits.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Designed for development and small-scale usage.	Deploy the API on cloud infrastructure and use load balancing for larger traffic

2	Data Storage	MongoDB is used for storing users and FAQs.	Use MongoDB Atlas and optimize indexes for larger datasets.
3	Performance	Requests are processed through a single backend server.	Introduce caching, optimized database queries, and scalable server deployment.
4	Security	JWT authentication, bcrypt password hashing, and protected routes are implemented.	Add stronger security controls, rate limiting, monitoring, and improved access management.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Cloud deployment and MongoDB Atlas integration	Short Term	Make the API accessible and easier to scale.
Phase 3	Semantic/vector-based FAQ search	Medium Term	Improve search results by understanding the meaning of user queries.
Phase 4	Admin dashboard and analytics	Long Term	Provide better FAQ management, usage monitoring, and system insights.